PURBANCHAL UNIVERSITY

2026

Master of Computer Application (M.C.A.)/First Semester/*Final*

Time: 03:00 hrs. Full Marks: 80 /Pass Marks: 32

MCA101: Discrete Structures *(New Course)*

MCA111: Descrete Mathematical Structures *(Old Course)*

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Group A

Answer TWO questions. **2×12=24**

1(a) Explain group homomorphism with an example. Show that the group G is abelian if and only if $(ab)^2=a^2b^2$. 3+7

(b) Prove that (Z, +) is a group where Z is a set of integers and + is an addition operation 6

2(a) Find an explicit formula for the fibonacci sequence 8

$f_n = f_{n-1} + f_{n-2}$ *for* $n >= 2$ with initial conditions $f_0 = 0$ *and* $f_1 = 1$.

(b) Prove that n^3-n is divisible by 3 whenever n is a positive integer using mathematical induction. 8

3(a) Let A= { 1, 2, 3, 4}.The relation R on A is given as:

R{(1, 1),(1,2),(2, 1),(2,2),(3, 1),(3,3),(1,3),(4.1),(4,4)}

(i) Construct the digraph of R. 3

(ii) Obtain the matrix of relation R. 3

(iii) List the in degrees and out degrees of the relation R. 3

(b) What is tautology and contingency? Explain with the help of a truth table. Explain predicate calculus with an example. 4+3

Group B

Answer SEVEN questions. **7×8=56**

4. Define an equivalence relation. Determine whether the relation a<=b on the set A={ 1,2,3,4,5 } is an equivalence relation or not.

Contd. ...

5. Discuss universal and existential quantifier. In how many different ways can the letters of the word 'ELECTION' be arranged so that the vowels always come together?

6. If a and b are positive integers such that a=252 and b=198. If d=GCD (a, b) then express d=sa+ tb for some integers s and t.

7. Determine the validity of the following argument by constructing the truth table:

If the economy of Nepal is poor, then the education system in Nepal will be poor. The education system in Nepal is poor.Therefore,the economy of Nepal is poor.

8. Define Boolean algebra. Explain POS and SOP of a Boolean function. What is canonical form of a Boolean function?

9. Define Euler Path and Euler Circuit. Use Fleury's Algorithm to find the Euler Circuit from the given graph.

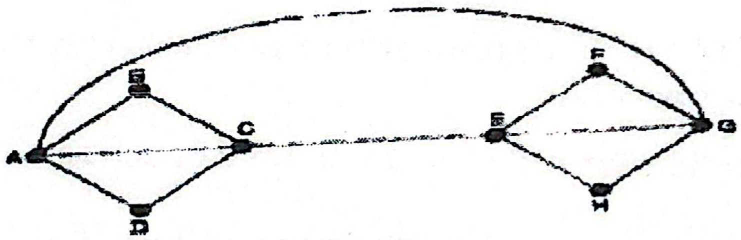
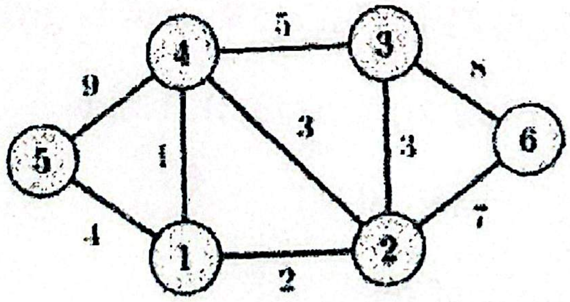


Fig. (1)

10. Give the definition of minimal spanning trees. Use Prim's algorithm to find the minimal spanning trees for the given. Graph. Consider '5' as the starting vertex.



11. What do you mean by transitive closure of R on A? Let A={1,2,3,4} and R={(1,2),(2,1),(2,3),(3,2),(3,4)}be the Relation on A. Then find the transitive closure of R using Warshall's Algorithm.

12. Write short notes on (Any TWO):

(a) Trees and its properties

(b) Planar and regular graphs

(c) Graph Isomorphism